

SAT Math Mastery · Week 9B

Capstone: Mixed Strategy & Timed Modules

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	mixed algebra, advanced math, data analysis and geometry · pick-a-number · back-solving · Desmos · eliminate by structure · pacing and flagging · the seven shared reasoning moves
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Classify a problem in the first ten seconds: algebra, advanced math, data, or geometry, and choose the cheapest method.
- S2.** Pick-a-number: test equivalent expressions, percent chains and “for all x ” statements with simple values.
- S3.** Back-solving: plug the answer choices into the original condition, starting with the easiest value to test.
- S4.** Desmos: intersections, sliders for parameters, tangency, tables for exponential and sequence questions.
- S5.** Eliminate by structure: signs, intercepts, end behavior, parity, units, domain, and “could NOT be” tests.
- S6.** Shared moves: structure and solution counts (M1), feature $\leftrightarrow$ form (M2), solve for a constant (M3), interpret (M4), scaling and units (M5), test a candidate (M6), extraneous and boundary care (M7).
- S7.** Run a timed two-pass module: answer, flag, return; never leave a question blank.
- S8.** Self-check habits: reread what is asked, check units and rounding, plug the answer back in.

2 · REQUIRED FACTS

- Digital SAT Math: 2 modules of 22 questions, 35 minutes each. The second module’s difficulty depends on the first; there is no penalty for a wrong answer.
- A system of two lines has no solution if the slopes are equal and the intercepts differ, and infinitely many if the two equations are the same line.
- A quadratic $ax^2 + bx + c$ with $a > 0$ has a minimum at $x = -\frac{b}{2a}$; its discriminant $b^2 - 4ac$ decides 2, 1 or 0 real solutions.
- $y = a \cdot b^t$ multiplies by b for every 1 unit of t ; $b > 1$ is growth and $0 < b < 1$ is decay.
- Mean is pulled toward outliers; median is not; spread (standard deviation) grows when values are farther from the mean.
- In a right triangle, $\sin \theta = \frac{\text{opp}}{\text{hyp}}$, $\cos \theta = \frac{\text{adj}}{\text{hyp}}$, $\tan \theta = \frac{\text{opp}}{\text{adj}}$; areas of similar figures scale by k^2 .
- Squaring both sides of an equation can create extraneous solutions: always test in the original equation.

3 · REQUIRED FORMULAS

$$y = mx + b, \quad (x - h)^2 + (y - k)^2 = r^2, \quad y = a(x - h)^2 + k, \quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{percent change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100, \quad \text{mean} = \frac{\text{sum}}{n}, \quad P(A | B) = \frac{\text{count in both}}{\text{count in } B}$$

$$\text{arc} = r\theta, \quad \text{sector area} = \frac{1}{2}r^2\theta = \frac{1}{2}rL, \quad a^2 + b^2 = c^2, \quad V_{\text{cyl}} = \pi r^2 h, \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h$$

4 · MINI THEORY SHEET

Pacing table (this 30-problem sheet = 90 minutes of solving).

Tier	Items	Target per item	Block time	Running clock at end of block
Easy	1–5	1 min	5 min	5 min
Medium	6–13	2 min	16 min	21 min
Hard	14–23	3 min	30 min	51 min
SAT Challenge	24–30	$5\frac{1}{2}$ min	38 min	89 min

On the real test one module averages $35 \div 22 \approx 1.6$ minutes per question. Rule: if you have spent twice the target time and have no plan, flag the question, mark a guess, and move on. Use the last 5 minutes of a module for flagged questions. A blank answer scores nothing; a guess can score.

Which method first?

Signal in the problem	First move
variables in the answer choices; “equivalent to”; “for all x ”	pick-a-number : test a simple value in the stem and in every choice
percent chain with no starting amount	pick 100 (or 10 and a ratio)
numeric choices and a messy equation	back-solve : plug a choice into the stem
intersection, “how many solutions”, a constant that makes a line tangent	Desmos : graph, add a slider for the constant
table or two points of an exponential or sequence	Desmos table or ratio of outputs
graph or sign conditions	eliminate by structure : sign of a , intercepts, vertex, end behavior
“could NOT be”; “must be true”	test candidates against every condition; one counterexample kills a statement
units, “nearest whole”, “at least”	convert first; round in the direction that satisfies the condition

Eliminate by structure. Before computing, ask which choices have the wrong sign, the wrong intercept, the wrong units, the wrong degree or the wrong order of magnitude. Cross them out first.

Pick-a-number rules. Avoid 0, 1 and numbers that make two choices agree; check *every* choice, because two can match one value.

Back-solving rules. Start with the choice that is easiest to compute; if it is too big or too small, you know which way to go.

Desmos rules. Type each equation exactly as written; use a slider for any constant; use the intersection points; check that the window shows every intersection.

The seven shared moves.

Move	Typical question
M1 count of solutions $\Leftrightarrow$ structure	equal slopes, discriminant = 0, same line
M2 feature $\leftrightarrow$ form	vertex form for the vertex, factored for the roots
M3 solve for a constant	use a given point or condition to find k
M4 interpret, don’t compute	meaning of a slope, intercept or residual
M5 scaling and units	squares, cubes, rates, conversions
M6 test a candidate against all conditions	inequality systems, tables, “which could be”
M7 extraneous and boundary care	radicals, whole-number rounding, strict inequalities

5 · GUIDED EXAMPLES

Example 1 — Pick-a-number (equivalent expressions)

For $x \neq 2$, which expression is equivalent to $\frac{3x^2 - x - 10}{x - 2}$? (A) $3x + 5$ (B) $3x - 5$ (C) $3x + 2$ (D) $3x - 10$

Pick $x = 3$: the stem gives $\frac{27 - 3 - 10}{1} = 14$. Choices: (A) 14, (B) 4, (C) 11, (D) -1 . Only (A) matches.
Algebra confirms it: $3x^2 - x - 10 = (3x + 5)(x - 2)$.

Example 2 — Back-solving (radical equation)

Which value of x satisfies $\sqrt{x + 9} = x - 3$? (A) 0 (B) 4 (C) 7 (D) 9

Test (C): $\sqrt{7 + 9} = 4$ and $7 - 3 = 4$ ✓. Test (A): $\sqrt{9} = 3$ but $0 - 3 = -3 \neq 3$ (extraneous). (B): $\sqrt{13} \neq 1$. (D): $\sqrt{18} \neq 6$. Answer: (C).

Example 3 — Desmos with a slider (tangency)

The line $y = kx - 7$ touches the parabola $y = x^2 - 2x - 3$ at exactly one point, and $k > 0$. Find k .

Graph the parabola and $y = kx - 7$ with a slider for k ; they touch when the discriminant of $x^2 - (2 + k)x + 4 = 0$ is 0: $(2 + k)^2 - 16 = 0 \Rightarrow k = 2$ or $k = -6$.

With $k > 0$: $k = 2$. (The touch point is $x = 2$, $y = -3$.)

6 · PRACTICE SET — 41 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

If $4x - 7 = 21$, what is the value of $4x + 7$?

(A) 28

(B) 35 ✓

(C) 7

(D) 14

1. Add 14 to both sides of the stem equation: $4x - 7 + 14 = 21 + 14$.

2. $4x + 7 = 35$. (Check: $4x = 28$, so $x = 7$ and $4(7) + 7 = 35$.)

Answer: (B)

2.

EASY · CALCULATOR

The function f is defined by $f(x) = 2x^2 - 3x + 1$. What is the value of $f(-2)$?

1. $f(-2) = 2(-2)^2 - 3(-2) + 1$.

2. $= 2(4) + 6 + 1 = 8 + 6 + 1 = 15$.

Answer: 15

3.

EASY · NO CALCULATOR NEEDED

The mean of five numbers is 12. Four of the numbers are 8, 10, 14 and 15. What is the fifth number?

1. Sum of all five $= 5 \cdot 12 = 60$.

2. Sum of the four given numbers $= 8 + 10 + 14 + 15 = 47$.

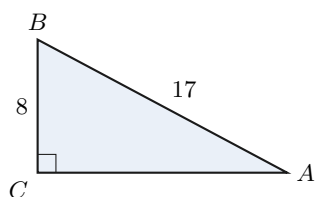
3. Fifth number $= 60 - 47 = 13$.

Answer: 13

4.

EASY · NO CALCULATOR NEEDED

In right triangle ABC shown, $\angle C = 90^\circ$, $AB = 17$ and $BC = 8$. What is the value of $\sin A$?

(A) $\frac{8}{15}$ (B) $\frac{17}{8}$ (C) $\frac{8}{17}$ ✓(D) $\frac{15}{17}$

1. $\sin A = \frac{\text{side opposite } A}{\text{hypotenuse}}$.

2. The side opposite $\angle A$ is $BC = 8$ and the hypotenuse is $AB = 17$.

3. $\sin A = \frac{8}{17}$.

Answer: (C)

5.

EASY · NO CALCULATOR NEEDED

For $x \neq -2$, which of the following expressions is equivalent to $\frac{2x^2 + x - 6}{x + 2}$?

(A) $2x - 3$ ✓

(B) $2x + 3$

(C) $2x - 6$

(D) $x - 3$

1. Factor the numerator: $2x^2 + x - 6 = (2x - 3)(x + 2)$.

2. Cancel $(x + 2)$: $\frac{(2x - 3)(x + 2)}{x + 2} = 2x - 3$.

3. (Pick $x = 1$: the stem gives $\frac{-3}{3} = -1$, and only $2x - 3$ equals -1 .)

Answer: (A)

6.

MEDIUM · NO CALCULATOR NEEDED

$$3x - 2y = 10$$

$$kx + 4y = 7$$

In the system of equations above, k is a constant. If the system has no solution, what is the value of k ?

(A) -3

(B) 6

(C) $\frac{3}{2}$

(D) -6 ✓

1. No solution means the lines are parallel and distinct: the slopes are equal.

2. Slopes: $\frac{3}{2}$ and $-\frac{k}{4}$. Set equal: $-\frac{k}{4} = \frac{3}{2} \Rightarrow k = -6$.

3. The right-hand sides are not in the same ratio ($\frac{10}{7} \neq -\frac{1}{2}$), so the lines are distinct and there is no solution.

Answer: (D)

7.

MEDIUM · CALCULATOR

A theater sold 40 tickets for a show. Adult tickets cost \$12 each and student tickets cost \$8 each. The total sales were \$392. How many adult tickets were sold?

(A) 12

(B) 16

(C) 18 ✓

(D) 22

1. Let a be the number of adult tickets, so $40 - a$ are student tickets.

2. $12a + 8(40 - a) = 392 \Rightarrow 4a + 320 = 392 \Rightarrow a = 18$.

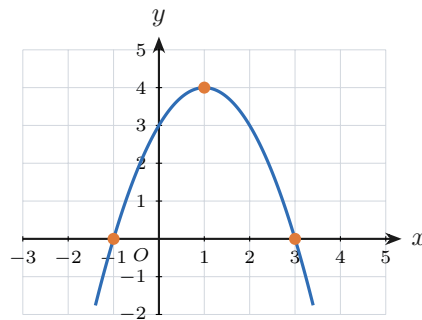
3. (Back-solve: $a = 18$ gives $216 + 176 = 392$ ✓.)

Answer: (C)

8.

MEDIUM · DESMOS

The graph of a quadratic function in the xy -plane is shown. Which of the following could be an equation of the graph?



(A) $y = (x - 1)^2 - 4$

(B) $y = -(x - 1)^2 + 4$ ✓

(C) $y = -(x + 1)^2 + 4$

(D) $y = -(x - 4)^2 + 1$

1. The graph opens downward, so the coefficient of the squared term is negative. This eliminates $y = (x - 1)^2 - 4$.
2. The vertex is $(1, 4)$, so the form is $y = -(x - 1)^2 + 4$.
3. Check an x -intercept: at $x = 3$, $-(2)^2 + 4 = 0$ ✓. The other choices fail at the vertex or the intercepts.

Answer: (B)

9.

MEDIUM · NO CALCULATOR NEEDED

A bacteria culture starts with 120 cells and doubles every 6 hours. The number of cells t hours after the start is modeled by $N(t) = 120 \cdot 2^{t/6}$. After how many hours will the culture have 1920 cells?

1. Set $120 \cdot 2^{t/6} = 1920$, so $2^{t/6} = 16$.
2. $16 = 2^4$, so $\frac{t}{6} = 4$.
3. $t = 24$.

Answer: 24

10.

MEDIUM · CALCULATOR

A survey asked 200 students whether they prefer online or in-person classes. The results are shown in the table.

	Online	In-person	Total
Grade 11	48	32	80
Grade 12	66	54	120
Total	114	86	200

If one of the students who prefer in-person classes is selected at random, what is the probability that the student is in Grade 12?

(A) $\frac{9}{20}$

(B) $\frac{27}{100}$

(C) $\frac{3}{5}$

(D) $\frac{27}{43}$ ✓

1. Restrict to the in-person column: $32 + 54 = 86$ students.

2. Grade 12 within that column: 54.

3. $P = \frac{54}{86} = \frac{27}{43}$.

Answer: (D)

11.

MEDIUM · NO CALCULATOR NEEDED

The value of p is 40% greater than the value of q , and the value of q is 25% less than the value of r , where $r > 0$. The value of p is what percent of the value of r ?

1. Pick $r = 100$. Then $q = 100 - 25 = 75$.

2. $p = 75 + 0.40(75) = 105$.

3. $\frac{p}{r} = \frac{105}{100} = 105\%$.

Answer: 105

12.

MEDIUM · NO CALCULATOR NEEDED

The equation $x^2 + y^2 - 6x + 4y = 12$ defines a circle in the xy -plane. What is the radius of the circle?

(A) 5 ✓

(B) 25

(C) $\sqrt{17}$

(D) 12

1. Complete the square: $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4$.

2. $(x - 3)^2 + (y + 2)^2 = 25$.

3. $r^2 = 25$, so $r = 5$.

Answer: (A)

13.

MEDIUM · NO CALCULATOR NEEDED

In right triangle ABC , $\angle B = 90^\circ$, $\sin A = \frac{3}{5}$ and $AC = 20$. What is the length of $\overline{BC}$?

(A) 25

(B) 12 ✓

(C) 16

(D) 15

1. $\overline{AC}$ is the hypotenuse. $\sin A = \frac{BC}{AC} = \frac{3}{5}$.

2. $BC = \frac{3}{5} \cdot 20 = 12$.

3. (The other leg is $AB = 16$, since $12^2 + 16^2 = 20^2$.)

Answer: (B)

14.

HARD · NO CALCULATOR NEEDED

$$(a - 2)x + 3 = 5x + b$$

In the equation above, a and b are constants. If the equation has infinitely many solutions for x , what is the value of $a + b$?

1. Infinitely many solutions means the two sides are identical expressions in x .

2. Match the x -coefficients: $a - 2 = 5 \Rightarrow a = 7$. Match the constants: $3 = b$.

3. $a + b = 7 + 3 = 10$.

Answer: 10

15.

HARD · NO CALCULATOR NEEDED

$$y < -x + 7$$

$$y > 2x - 5$$

Which of the following ordered pairs (x, y) is a solution to the system of inequalities above?

(A) $(1, 6)$ (B) $(0, -6)$ **(C) $(3, 3)$ ✓**(D) $(4, 2)$

1. Test each pair in both inequalities.

2. $(3, 3)$: $3 < -3 + 7 = 4$ ✓ and $3 > 2(3) - 5 = 1$ ✓.

3. $(4, 2)$: $2 > 3$ is false. $(1, 6)$: $6 < 6$ is false (strict). $(0, -6)$: $-6 > -5$ is false.

Answer: (C)

16.

HARD · DESMOS

$$y = x^2 - 6x + c$$

$$y = 2x - 7$$

In the system above, c is a constant. If the system has exactly one solution, what is the value of c ?

1. Set the expressions equal: $x^2 - 6x + c = 2x - 7 \Rightarrow x^2 - 8x + (c + 7) = 0$.

2. Exactly one solution means the discriminant is 0: $(-8)^2 - 4(1)(c + 7) = 0$.

3. $64 = 4(c + 7) \Rightarrow c + 7 = 16 \Rightarrow c = 9$.

Answer: 9

17.

HARD · NO CALCULATOR NEEDED

What is the sum of all solutions to the equation $\sqrt{3x+4} = x$?

(A) 4 ✓

(B) 5

(C) -1

(D) 3

1. Square both sides: $3x + 4 = x^2 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0$.
2. Test $x = 4$: $\sqrt{16} = 4$ ✓. Test $x = -1$: $\sqrt{1} = 1 \neq -1$, so $x = -1$ is extraneous.
3. The only solution is 4, so the sum is 4.

Answer: (A)

18.

HARD · CALCULATOR

The function $f(t) = a \cdot b^t$, where a and b are positive constants, satisfies $f(1) = 12$ and $f(4) = 324$. What is the value of $f(6)$?

(A) 8748

(B) 972

(C) 1458

(D) 2916 ✓

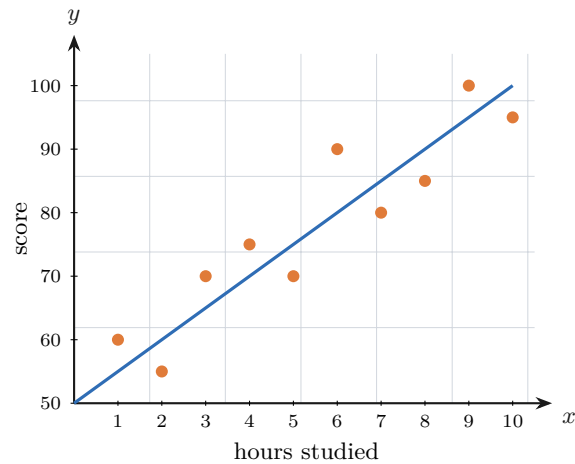
1. $\frac{f(4)}{f(1)} = b^3 = \frac{324}{12} = 27$, so $b = 3$.
2. $f(1) = 3a = 12 \Rightarrow a = 4$.
3. $f(6) = 4 \cdot 3^6 = 4 \cdot 729 = 2916$.

Answer: (D)

19.

HARD · DESMOS

The scatterplot shows the number of hours studied, x , and the test score, y , for 10 students. The line of best fit shown passes through $(2, 60)$ and $(8, 90)$. The residual of a data point is its actual score minus the score predicted by the line. For which number of hours studied is the absolute value of the residual greatest?



(A) 10

(B) 2

(C) 6 ✓

(D) 9

- Slope of the line: $\frac{90 - 60}{8 - 2} = 5$; $y = 5x + 50$.
- Residual = actual - predicted: $x = 1$: $60 - 55 = 5$; $x = 2$: $55 - 60 = -5$; $x = 3$: $70 - 65 = 5$; $x = 4$: $75 - 70 = 5$; $x = 5$: $70 - 75 = -5$.
- $x = 6$: $90 - 80 = 10$; $x = 7$: $80 - 85 = -5$; $x = 8$: $85 - 90 = -5$; $x = 9$: $100 - 95 = 5$; $x = 10$: $95 - 100 = -5$.
- The largest absolute residual is 10, at 6 hours.

Answer: (C)

20.

HARD · NO CALCULATOR NEEDED

Data set A is 2, 4, 6, 8, 10. Data set B is 2, 4, 6, 8, 40. Which of the following statements is true?

- I. The mean of B is greater than the mean of A .
- II. The median of B is greater than the median of A .
- III. The standard deviation of B is greater than the standard deviation of A .

(A) I only

(B) I and III only ✓

(C) II and III only

(D) I, II, and III

- 1. I: mean of $A = 6$, mean of $B = \frac{60}{5} = 12$. True.
- 2. II: both medians are 6. False.
- 3. III: the value 40 is far from the mean, so B is much more spread out than A . True.
- 4. Statements I and III only.

Answer: (B)**21.**

HARD · CALCULATOR

A machine fills 15 bottles every 40 seconds. At this rate, how many hours will it take the machine to fill 4050 bottles?

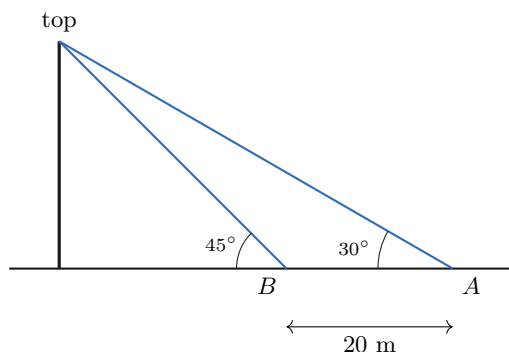
- 1. Number of 40-second cycles: $\frac{4050}{15} = 270$.
- 2. Time = $270 \cdot 40 = 10,800$ seconds.
- 3. $\frac{10,800}{3600} = 3$ hours.

Answer: 3

22.

HARD · CALCULATOR

From point A on level ground, the angle of elevation to the top of a vertical tower is 30° . From point B , which is 20 meters closer to the base of the tower and on the same line, the angle of elevation is 45° . What is the height of the tower, in meters?



(A) 20

(B) $10(\sqrt{3} - 1)$ (C) $10\sqrt{3}$ **(D) $10(\sqrt{3} + 1)$ ✓**

- Let the height be h and let d be the distance from B to the base. At B : $\tan 45^\circ = \frac{h}{d} = 1$, so $d = h$.
- At A : $\tan 30^\circ = \frac{h}{h+20} = \frac{1}{\sqrt{3}} \Rightarrow \sqrt{3}h = h + 20$.
- $h(\sqrt{3} - 1) = 20 \Rightarrow h = \frac{20}{\sqrt{3} - 1} = \frac{20(\sqrt{3} + 1)}{2} = 10(\sqrt{3} + 1) \approx 27.3$.

Answer: (D)**23.**

HARD · NO CALCULATOR NEEDED

In triangle ABC , $AB = AC = 15$ and $BC = 24$. Point D lies on $\overline{AC}$ such that $\overline{BD} \perp \overline{AC}$. What is the length of $\overline{BD}$?

- Drop the altitude from A to $\overline{BC}$: it bisects $\overline{BC}$, so $h = \sqrt{15^2 - 12^2} = 9$.
- Area of $\triangle ABC = \frac{1}{2}(24)(9) = 108$.
- Using $\overline{AC}$ as the base: $\frac{1}{2}(15)(BD) = 108 \Rightarrow BD = \frac{216}{15} = 14.4$.

Answer: 14.4

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$(k + 1)x + 6y = 18$$

$$4x + (k - 1)y = 12$$

In the system above, k is a constant. For which value of k does the system have no solution?

1. The determinant condition for parallel lines: $(k + 1)(k - 1) = 6 \cdot 4 \Rightarrow k^2 - 1 = 24 \Rightarrow k = 5$ or $k = -5$.
2. $k = 5$: the equations are $6x + 6y = 18$ and $4x + 4y = 12$; both reduce to $x + y = 3$, the same line, so infinitely many solutions.
3. $k = -5$: $-4x + 6y = 18$ and $4x - 6y = 12$. Adding gives $0 = 30$, which is impossible, so no solution.
4. The value is $k = -5$.

Answer: -5 **25.**

SAT CHALLENGE · NO CALCULATOR NEEDED

A jar contains only red, blue and green marbles. The ratio of red to blue marbles is $3 : 5$, and the ratio of blue to green marbles is $2 : 3$. There are 36 more green marbles than red marbles. How many marbles are in the jar?

1. Make the blue part match: red : blue = $3 : 5 = 6 : 10$ and blue : green = $2 : 3 = 10 : 15$.
2. So red : blue : green = $6 : 10 : 15$. Let the parts be $6u, 10u, 15u$.
3. Green - red = $15u - 6u = 9u = 36 \Rightarrow u = 4$.
4. Total = $(6 + 10 + 15)u = 31 \cdot 4 = 124$.

Answer: 124

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function $f(x) = x^2 + bx + c$ has a minimum value of -25 , and $f(0) = -21$. What is the greatest possible value of b ?

1. $f(0) = c = -21$.

2. The minimum value is $c - \frac{b^2}{4}$ (at $x = -\frac{b}{2}$): $-21 - \frac{b^2}{4} = -25 \Rightarrow b^2 = 16$.

3. $b = 4$ or $b = -4$; the greatest possible value is 4.

Answer: 4

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

For what value of x is $2^{x+3} - 2^x = 56$?

(A) 3 ✓

(B) 4

(C) 7

(D) 8

1. Factor: $2^{x+3} - 2^x = 2^x(2^3 - 1) = 7 \cdot 2^x$.

2. $7 \cdot 2^x = 56 \Rightarrow 2^x = 8$.

3. $x = 3$. (Check: $2^6 - 2^3 = 64 - 8 = 56$.)

Answer: (A)

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

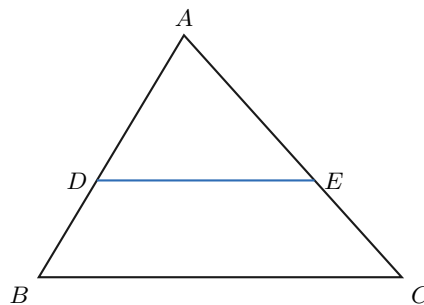
The mean of a list of 9 numbers is 20. Two numbers are removed from the list; one of them is twice the other. The mean of the remaining 7 numbers is 18. What is the larger of the two numbers that were removed?

1. Sum of the original list = $9 \cdot 20 = 180$. Sum of the remaining 7 = $7 \cdot 18 = 126$.
2. Sum of the two removed numbers = $180 - 126 = 54$.
3. Let them be x and $2x$: $3x = 54 \Rightarrow x = 18$.
4. The larger number is $2x = 36$.

Answer: 36**29.**

SAT CHALLENGE · NO CALCULATOR NEEDED

In triangle ABC , point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$, as shown. $AD = 6$, $DB = 4$, and the area of triangle ADE is 18. What is the area of quadrilateral $DBCE$?



1. $AB = AD + DB = 10$. Since $\overline{DE} \parallel \overline{BC}$, $\triangle ADE \sim \triangle ABC$ with ratio $\frac{AD}{AB} = \frac{6}{10} = \frac{3}{5}$.
2. Areas scale by the square: $\frac{[ADE]}{[ABC]} = \frac{9}{25}$.
3. $[ABC] = 18 \cdot \frac{25}{9} = 50$.
4. $[DBCE] = 50 - 18 = 32$.

Answer: 32

30.

SAT CHALLENGE · CALCULATOR

A sector of a circle has area 20π and arc length 4π . What is the measure, in radians, of the central angle of the sector?

(A) $\frac{\pi}{5}$

(B) $\frac{2\pi}{5}$ ✓

(C) $\frac{4\pi}{5}$

(D) $\frac{\pi}{2}$

1. Sector area $= \frac{1}{2}rL$ where L is the arc length: $\frac{1}{2}r(4\pi) = 20\pi \Rightarrow r = 10$.

2. Arc length $L = r\theta \Rightarrow 4\pi = 10\theta$.

3. $\theta = \frac{4\pi}{10} = \frac{2\pi}{5}$.

Answer: (B)

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial function p is defined by $p(x) = -(x+3)^2(x-1)^3(x-4)^2$. Consider the following statements.

I. $p(x) \geq 0$ for all $x < 1$.

II. The graph of $y = p(x)$ crosses the x -axis at $x = 4$.

III. $p(x) \rightarrow \infty$ as $x \rightarrow -\infty$.

Which of the statements must be true?

(A) I only

(B) I and II only

(C) I and III only ✓

(D) II and III only

1. Multiplicities: $(x+3)^2$ and $(x-4)^2$ are even, so the graph touches the axis at -3 and 4 without crossing. $(x-1)^3$ is odd, so the graph crosses at 1 .

2. I: for $x < 1$, $(x-1)^3 < 0$, so $-(x-1)^3 > 0$, while the squares are ≥ 0 . Hence $p(x) \geq 0$, with $p(-3) = 0$. True. (The sign does not change at $x = -3$, but $p(-3) = 0$, so $\geq$ is needed.)

3. II: $x = 4$ has even multiplicity, so the graph touches and does not cross. False.

4. III: the degree is $2 + 3 + 2 = 7$ (odd) and the leading coefficient is negative, so $p(x) \rightarrow +\infty$ as $x \rightarrow -\infty$ (and $-\infty$ as $x \rightarrow \infty$). True.

5. I and III only.

Answer: (C)

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial $12x^4 - 84x^2 + 144$ can be written in the form $(ax^2 + b)(cx^2 + d)$, where a , b , c and d are integers with $a > 0$ and $c > 0$. What is the least possible value of $a + b + c + d$?

(A) -22 (B) -18 **(C) -38 ✓**(D) -27

- Factor out the common 12: $12(x^4 - 7x^2 + 12) = 12(x^2 - 3)(x^2 - 4)$. Neither $x^2 - 3$ nor $x^2 - 4$ can be rewritten with a different pair of roots, so every valid form is $\alpha(x^2 - 3) \cdot \beta(x^2 - 4)$ with $\alpha\beta = 12$.
- Then $a = \alpha$, $b = -3\alpha$, $c = \beta$, $d = -4\beta$, and $a + b + c + d = -2\alpha - 3\beta$.
- Test the positive pairs (α, β) with $\alpha\beta = 12$: $(1, 12) \rightarrow -38$; $(2, 6) \rightarrow -22$; $(3, 4) \rightarrow -18$; $(4, 3) \rightarrow -17$; $(6, 2) \rightarrow -18$; $(12, 1) \rightarrow -27$.
- The least is -38 , from $(x^2 - 3)(12x^2 - 48)$. Check: $(x^2 - 3)(12x^2 - 48) = 12x^4 - 48x^2 - 36x^2 + 144 = 12x^4 - 84x^2 + 144$ and $1 - 3 + 12 - 48 = -38$.
- (Moving the whole 12 onto the first binomial gives $(12x^2 - 36)(x^2 - 4)$, sum -27 , which is not the least.)

Answer: (C)**33.**

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive real number m , let $a = \sqrt[3]{m}$ and $b = \sqrt[4]{m}$. If $ab = 128$, what is the value of ab^2 ?

- Write both with the same base: $a = m^{1/3}$, $b = m^{1/4}$.
- $ab = m^{1/3+1/4} = m^{7/12} = 128 = 2^7$, so $m = 2^{12} = 4096$ (the function $m^{7/12}$ is increasing, so this is the only solution).
- $ab^2 = m^{1/3+1/2} = m^{5/6} = 2^{10} = 1024$.
- (Check: $a = 16$, $b = 8$, $ab = 128$, $ab^2 = 16 \cdot 64 = 1024$.)

Answer: 1024

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

Positive real numbers a and b , both greater than 1, satisfy $a^3 = b^2$. If $\frac{a^9}{\sqrt[3]{b}} = b^k$, what is the value of k ?

(A) $\frac{17}{3}$ ✓

(B) 6

(C) $\frac{19}{3}$

(D) $\frac{17}{2}$

1. From $a^3 = b^2$: $a^9 = (a^3)^3 = (b^2)^3 = b^6$.

2. $\sqrt[3]{b} = b^{1/3}$, so $\frac{a^9}{\sqrt[3]{b}} = \frac{b^6}{b^{1/3}} = b^{6-\frac{1}{3}} = b^{17/3}$.

3. Since $b > 1$, equal powers of b have equal exponents: $k = \frac{17}{3}$.

4. (Option 6 ignores the cube root; $\frac{19}{3}$ adds instead of subtracting $\frac{1}{3}$.)

Answer: (A)

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

The temperature T , in degrees Fahrenheit, of a sample t minutes after it is placed in a chamber is modeled by one of the functions below for $t \geq 0$. For which model is 85 the greatest temperature the sample actually reaches?

(A) $T = 85 - 60(0.8)^t$

(B) $T = 40 + 45(1.05)^t$

(C) $T = 25 + 85(0.8)^t$

(D) $T = 25 + 60(0.8)^t$ ✓

1. Test each model on $t \geq 0$ and ask whether 85 is reached, not merely approached.

2. The model $T = 85 - 60(0.8)^t$ increases toward 85 but is always below it: 85 is a horizontal asymptote, never attained. No maximum.

3. The model $T = 25 + 60(0.8)^t$ decreases, so its greatest value is at $t = 0$: $25 + 60 = 85$. Attained. ✓

4. The model $T = 40 + 45(1.05)^t$ grows without bound, and $T = 25 + 85(0.8)^t$ has maximum $25 + 85 = 110$ at $t = 0$.

5. Only $T = 25 + 60(0.8)^t$ has 85 as an attained greatest value.

Answer: (D)

36.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = \frac{6^{x+1} \cdot 4^x}{3^{x-2}}$. It can be written in the form $f(x) = a \cdot b^{x-1}$, where a and b are positive constants. What is the value of $a + b$?

(A) 3456

(B) 62

(C) 440 ✓

(D) 456

1. Evaluate at $x = 1$, where $b^{x-1} = 1$: $a = f(1) = \frac{6^2 \cdot 4}{3^{-1}} = 36 \cdot 4 \cdot 3 = 432$.

2. Find b from the per-unit multiplier: $\frac{f(x+1)}{f(x)} = \frac{6 \cdot 4}{3} = 8$, so $b = 8$.

3. Check: $f(x) = 6 \cdot 9 \cdot \frac{6^x 4^x}{3^x} = 54 \cdot 8^x = 432 \cdot 8^{x-1}$.

4. $a + b = 432 + 8 = 440$.

Answer: (C)

37.

SAT CHALLENGE · CALCULATOR

An investment grows by 6% every half-year. If V_0 is its initial value, the value after t years is $V(t)$. Which of the following gives the number of years t until the value is three times the initial value?

(A) $t = \log_{1.06} \frac{3}{2}$ (B) $t = \frac{1}{2} \log_{1.06} 3$ ✓(C) $t = 2 \log_{1.06} 3$ (D) $t = \log_{1.12} 3$

1. A 6% increase multiplies the value by 1.06 every half-year, so in t years there are $2t$ periods: $V(t) = V_0(1.06)^{2t}$.

2. Set $V(t) = 3V_0$ and divide by V_0 to cancel the constant: $(1.06)^{2t} = 3$.

3. Take $\log_{1.06}$ of both sides: $2t = \log_{1.06} 3$, so $t = \frac{1}{2} \log_{1.06} 3$.

4. (Numerically $t \approx 9.43$ years; $(1.06)^{18.86} \approx 3$.)

Answer: (B)

38.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$T(t) = 24 + 66(0.85)^t$$

The function T models the temperature, in degrees Celsius, of a cup of tea t minutes after it is placed in a room. Which of the following is the best interpretation of the number 0.85 in this context?

(A) T falls 15% per minute**(B) the gap $T - 24$ falls 15% per minute ✓**(C) $T(1) = 0.85T(0)$ (D) T levels off at 85% of $T(0)$

1. The constant 24 is the room temperature: the tea cools toward 24°C , not toward 0.
2. Subtract it: $T(t) - 24 = 66(0.85)^t$. Each minute this difference is multiplied by 0.85, so it decreases by 15%.
3. The tea's own temperature does not decrease by 15% per minute: $T(0) = 90$ and $T(1) = 80.1$, a drop of about 11%.
4. So 0.85 is the factor by which the difference between the tea and the room changes each minute.

Answer: (B)

39.

SAT CHALLENGE · NO CALCULATOR NEEDED

For all real numbers x , the function g is defined by $g(x) = \frac{9^{x+2} \cdot 10^{x+1}}{6^{x-1}}$. The function can be written in the form $g(x) = k \cdot 15^{x+1}$, where k is a constant. What is the value of k ?

1. Because $15^{x+1} = 1$ when $x = -1$, the constant k equals $g(-1)$.
2. $g(-1) = \frac{9^1 \cdot 10^0}{6^{-2}} = 9 \cdot 36 = 324$.
3. (Check by primes: $g(x) = \frac{3^{2x+4} 2^{x+1} 5^{x+1}}{2^{x-1} 3^{x-1}} = 3^{x+5} \cdot 2^2 \cdot 5^{x+1} = 4 \cdot 3^5 \cdot 5 \cdot 15^x = 4860 \cdot 15^x = 324 \cdot 15^{x+1}$.)

Answer: 324

40.

SAT CHALLENGE · CALCULATOR

The concentration of a medication in the bloodstream, in milligrams per liter, is modeled by $C(t) = 18 \left(\frac{1}{3}\right)^{\frac{t-4}{2}}$, where $t \geq 0$ is the number of hours after the first measurement. Which of the following is an equivalent form of $C(t)$ that displays the maximum value of C as a constant or coefficient?

(A) $54 \cdot 3^{\frac{2-t}{2}}$

(B) $486 \cdot 3^{-\frac{t+2}{2}}$

(C) $18 \cdot 3^{\frac{4-t}{2}}$

(D) $162 \cdot 3^{-\frac{t}{2}}$ ✓

- $\left(\frac{1}{3}\right)^{\frac{t-4}{2}} = 3^{\frac{4-t}{2}}$, so C is a decreasing function of t and its maximum on $t \geq 0$ occurs at $t = 0$.
- $C(0) = 18 \cdot 3^2 = 162$.
- Split the exponent: $18 \cdot 3^{\frac{4-t}{2}} = 18 \cdot 3^2 \cdot 3^{-\frac{t}{2}} = 162 \cdot 3^{-\frac{t}{2}}$. The coefficient is the value at $t = 0$, i.e. the maximum.
- The other options are equivalent too, but their exponents equal 2, 1 and -1 at $t = 0$, so their coefficients (18, 54, 486) are not the maximum.

Answer: (D)

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, a circle is tangent to both the x -axis and the y -axis, has its center in the second quadrant, and passes through the point $(-2, 9)$. The diameter of the circle is less than 15. The circle has equation $x^2 + y^2 + Dx + Ey + F = 0$, where D , E and F are constants. What is the value of F ?

- Tangent to both axes with center in the second quadrant: center $(-r, r)$, radius $r > 0$. Equation: $(x+r)^2 + (y-r)^2 = r^2$.
- Expand: $x^2 + y^2 + 2rx - 2ry + r^2 = 0$, so $F = r^2$.
- Point $(-2, 9)$: $(-2+r)^2 + (9-r)^2 = r^2 \Rightarrow r^2 - 22r + 85 = 0 \Rightarrow (r-5)(r-17) = 0$.
- Diameter < 15 means $r < 7.5$, so $r = 5$ and $F = 25$. (The root $r = 17$ gives diameter 34.)

Answer: 25